Sharing of Irreversible Latent Features in lieu of RF IQ Data for Multiple Downstream Learning Tasks

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Abstract

Wireless in-phase and quadrature (I/Q) data is useful for training various radio frequency (RF) machine learning (ML) models. However, sharing such I/Q data faces barriers from concerns of (i) potentially violating laws related to recording raw samples that can eventually be decoded by an unauthorized third party, and (ii) revealing information about a proprietary system or the environment involved in the data capture. This paper proposes a different solution to sharing I/Q data intended for use in ML models by leveraging latent embeddings obtained through a shared encoder. The key contributions include: (i) demonstrating how a single set of common features can be extracted from an I/Q dataset for training multiple downstream RF classification or regression tasks, (ii) designing a method of structured noise infusion that impairs the reverse recovery of the raw I/Q from the feature space without significant impact to the downstream tasks, and (iii) demonstrating the application of this method on informative real-world, publicly available datasets. Sharing well-constructed yet privacy-preserving features can accelerate foundational research on ML for wireless without relying on hard-to-get I/Q samples.

CCS Concepts

• **Networks** → **Wireless access networks**; • **Computing methodologies** → *Machine learning*; • **Hardware** → *Wireless devices.*

Keywords

Wireless communications, I/Q datasets, representation learning, intermediate feature vectors, multi-task learning, irreversible feature extraction

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1 Introduction

Despite great strides within the ML community in the domains of natural language processing, image analysis, text generation, among others, the adoption of ML for physical-layer wireless use cases has been slow. Several governmental findings as well as industry-led reports have pointed towards the root cause as the lack of available datasets for training ML models and the lack of incentives for sharing such datasets with third parties. While in the case of natural images, datasets can be collected and shared with considerably fewer restrictions, sharing RF datasets raises several concerns. In the US, the key concern is whether recording, storing, and sharing RF I/Q samples violates the Electronic Communications Privacy Act of 1986 (ECPA) that *prohibits the intentional actual interception of any wire, oral, or electronic communication* [4]. Sharing feature vectors that are encoded I/Q data can be one solution to address this privacy concern, however, this leaves the issue of reverse-engineering and recovering at least a part of user I/Q data unaddressed. Thus, without a radical change in the way we store and share RF datasets, and use them to train models the barriers towards adopting robust and high-value ML models at industry level cannot be overcome.

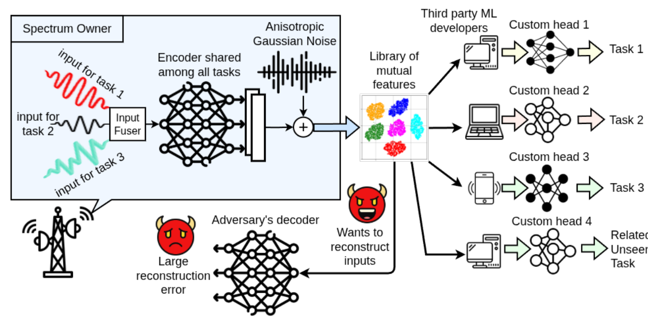

Figure 1: An overview of the proposed approach where a shared encoder trained on multiple downstream tasks generates privacy-preserving features that third-party ML developers can use to train task-specific heads, without access to raw I/Q data.

• **Approach.** As shown in Fig. 1, our key intuition is to split an end-to-end model into an *encoder* for feature extraction and potential third-party *custom head(s)*. Our approach enables the spectrum owner to create and share libraries of higher-dimensional latent embeddings or features and share them instead of raw I/Q samples. These features are extracted from raw I/Q sequences mutually for multiple tasks through a carefully designed neural network encoder. They capture the inherent properties of the inputs that can potentially inform multiple downstream tasks. For an RF signal, the features may embed different signal impairments including carrier frequency offset (CFO), impact of the wireless channel, hardware-specific signal distortions, or inherent signal properties such as effects of specific modulation schemes, among others. Thus, when

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these features are provided to third party ML developers, they can train their *custom head(s)* for different classification or regression tasks without accessing the raw I/Q data (see Fig. 1).

• **Challenges.** While generating intermediate features is conceptually straightforward, there are several issues to consider. First, there could be multiple downstream tasks, and the same common feature set should be used for all of them, with the objective of reducing the overhead of generating separate features for each task **(C1)**. Second, the features are expected to be shared externally (in lieu of I/Q). Assuming that a subset of training samples is leaked, allowing an adversary to query the encoder and obtain corresponding latent features, it should still be infeasible for a third party to recover the original I/Q samples from these (data, feature) pairs **(C2)**. Third, the process of making it difficult for an adversary to reconstruct inputs should result in minimal degradation of utility, i.e., the released features should remain competitive with end-to-end models on downstream tasks **(C3)**.

We address these challenges by adapting techniques from *multi-task representation learning* and applying *information-theoretic principles* for data obfuscation. Interestingly, multi-task learning naturally creates an information bottleneck, forcing the model to learn a generalized representation of the input data [15, 21, 28]. This process inherently tries to discard non-essential information, which not only improves model generalization but also complicates the reconstruction of the original input $\mathbf{x}$ from the latent feature vector $\mathbf{z}$. In addition to this inherent property of multi-task learning, we introduce a post-hoc task-aware anisotropic noise injection mechanism to further distort the generated features based on the learned tasks and make them **hard** to reverse back to the original I/Q sequences.

We note that our scope and focus is different from foundation model design based training across a large and diverse body of data, where the goal is to generalize to unseen wireless channels [2], or unseen tasks [19, 33], as such features might still compromise user privacy by being reversible back to user I/Q data. In contrast, once trained, our encoder representations are specific to the learned tasks, but since it is based on multitask learning, it is also expected to perform well on new tasks that mostly reside in the lower-dimension shared subspace of tasks it was trained on [20], and the corresponding sample complexity will depend on the dimension of this shared representation [27]. Of course if a new task warrants a very different data representation, our approach will not provide such guarantees. We deliberately do not try to achieve arbitrary cross-task generalization and "zero-shot" generalization as aimed for in foundation models, since data privacy is an important constraint in our setting. We do empirically show that representations transfer to semantically related unseen tasks (§ 6.8). To demonstrate the proposed framework, we use a publicly available over-the-air dataset and define three tasks as representative RF examples: transmitter classification (i.e., RF fingerprinting), channel estimation, and coarse CFO estimation. The proposed scheme for multi-task learning extracting irreversible features achieves a 12% RF fingerprinting accuracy gain from joint learning while incurring only a 5% channel estimation utility degradation. Furthermore, the proposed anisotropic noise injection mechanism outperforms the isotropic noise baseline, which suffers from a 94% degradation in downstream task utility (§ 6.4).

Our key contributions are as follows.

- In § 3, we introduce a joint learning framework that allows third-party ML developers to train models for diverse wireless communication tasks without direct access to the raw user data.
- In § 4, we propose an anisotropic noise injection mechanism and provide theoretical guarantees showing that the injected noise makes data reconstruction by adversaries significantly more difficult. To the best of our knowledge, this is the first task-aware noise injection method specifically designed for multi-task learning of RF problems.
- In § 5, we describe three typical wireless use cases of RF fingerprinting, channel estimation, and coarse CFO estimation that rely on ML models trained with a publicly available, over-the-air collected WiFi dataset (§ 5.1). Our exhaustive performance evaluations are presented in § 6.
- We make the feature extractor software framework available as a community resource, to allow widespread data collection and sharing efforts[1] [1].

2 Related Work

I/Q data compression. Several work reduce the cost of storing or transmitting raw I/Q data through learned compression. [29] uses attention-based asymmetric CNNs, [22] applies vector quantization across six modulation classes, and [3] employs singular value decomposition for low-rank time-frequency approximations. These work compress I/Q samples to save bandwidth and storage, whereas our approach shares features rather than raw I/Q, with the explicit goal of preserving privacy.

Feature extraction for RF tasks. [24] trains a neural feature extractor with triplet loss on LoRa preambles for out-of-distribution device identification. [14] uses an Extra Trees wrapper to derive reduced feature vectors for a wireless intrusion detection system, demonstrating that cascading a feature extractor with a deep classifier outperforms end-to-end training. [8] employs distributed ML for downlink channel estimation, training joint feature extractors at multiple users with a shared classifier at the base station. None of these work address multi-task sharing or privacy.

Privacy-preserving data release. The field traces back to *k-anonymity* [25] and was transformed by *differential privacy* (DP) [9], which remains the theoretical gold standard. Variants including Rényi-DP, local-DP, and metric-DP followed, but DP is often seen to be impractical for deep networks [11].

Learnable encodings with privacy guarantees. *Instahide* [12] gained attention but was broken [5]. *Neuracrypt* [32] and subsequent learnable encryption methods share encoded data in lieu of raw inputs. More recent work derives formal non-invertibility guarantees via the Fisher information matrix [18], conditional entropy bounds [30], and Hammersley-Chapman-Robbins (HCR) lower bounds [6]. PAC Privacy [31] unifies several obfuscation mechanisms under one theoretical framework. Our work differs in two key ways: we focus on *post-training* plug-and-play mechanisms, and we provide formal guarantees in a *multi-task learning* (MTL) setting.

[1]The final version of the code and model checkpoints will be released publicly upon acceptance of the paper

3 Problem Scope and Objectives

3.1 Scenario Description

Let $\mathcal{T} = \{T_1, T_2, \ldots, T_N\}$ denote a collection of downstream learning tasks (we specifically consider the tasks of modulation classification, RF fingerprinting, or parameter estimation such as channel or CFO estimation in this work). For each task T_i, we are given a dataset of M_i samples $\mathcal{D}_i = \{(\mathbf{x}_{i,j}, \mathbf{y}_{i,j})\}_{j=1}^{M_i}$ where $\mathbf{x}_{i,j} \in \mathcal{X}_i \subseteq \mathbb{R}^{d_{x_i}}$ is an input vector (e.g., a user I/Q sequence that serves as an input sample to the ML model) and $\mathbf{y}_{i,j} \in \mathcal{Y}_i \subseteq \mathbb{R}^{d_{y_i}}$ is the associated label vector. A third-party ML developer desires to train task-specific models using data curated by a spectrum manager. The latter has originally collected the $\{\mathcal{D}_i\}_{i=1}^{N}$ and wishes to enable training for any subset of tasks $S \subseteq \mathcal{T}$ by the ML developer. We assume that releasing labels $\mathbf{y}_{i,j}$ is permissible, whereas releasing raw inputs $\mathbf{x}_{i,j}$ is not. A possible alternative is to share intermediate feature representations produced by a trained encoder instead of raw $\mathbf{x}$. However, if an adversary gains access to even a small subset of I/Q sequences together with their features, a decoder can be trained to reconstruct inputs with high fidelity, thereby compromising privacy. This motivates two interconnected challenges:

Problem 1 (Shared Representation Learning). Learn and share a representation of inputs that is broadly useful across multiple tasks, mitigating the need to distribute each full dataset $\mathcal{D}_i$.

Problem 2 (Privacy Preservation). Ensure that the shared representations do not permit accurate reconstruction of private inputs $\mathbf{x}_{i,j}$ by an adversary while preserving task utility.

3.2 Formal Definition

We define the *universe of inputs* across all tasks as: $\mathcal{X}_\cup = \bigcup_{i=1}^{N} \mathcal{X}_i$. Let $\mathcal{P}(\mathcal{T})$ denote the power set of $\mathcal{T}$. For any subset $S \in \mathcal{P}(\mathcal{T})$ with cardinality $|S| = r$, define the *joint input domain* $\mathcal{X}_S = \prod_{T_i \in S} \mathcal{X}_i$, so that an element of $\mathcal{X}_S$ is a tuple $\mathbf{x}_{1:r} = (\mathbf{x}_{t_1}, \ldots, \mathbf{x}_{t_r})$ with $\mathbf{x}_{t_\ell} \in \mathcal{X}_{t_\ell}$. For $r = 1$ this reduces to the single-task domain $\mathcal{X}_i$.

3.2.1 Encoder and task heads. A *pretrained deterministic encoder*, F, is a mapping between the joint input domain $\mathcal{X}_S$ to a latent representation space as shown in (1).

$$F : \bigcup_{S \in \mathcal{P}(\mathcal{T})} \mathcal{X}_S \longrightarrow \mathcal{Z} \subseteq \mathbb{R}^k, \qquad \mathbf{z} = F(\mathbf{x}_{1:r}) \tag{1}$$

F transforms an input tuple into a latent representation $\mathbf{z}$. For each task T_i, a corresponding *task head* $G_i : \widehat{\mathcal{Z}} \to \mathcal{Y}_i$ produces predictions $\widehat{\mathbf{y}}_i = G_i(\widehat{\mathbf{z}})$ from released features $\widehat{\mathbf{z}} \in \widehat{\mathcal{Z}}$ (defined below). Here, $k \in \mathbb{N}$ is the latent dimension, and all spaces and mappings are assumed measurable.

3.2.2 Adversary model. Let $\mathcal{D}_{\text{leak}} \subseteq \bigcup_{i=1}^{N} \mathcal{D}_i$ be a compromised subset of data that is available to the adversary. We define $\mathcal{X}_{\text{leak}} = \{\mathbf{x}_{i,j} \mid (\mathbf{x}_{i,j}, \mathbf{y}_{i,j}) \in \mathcal{D}_{\text{leak}}\}$ and $\mathcal{Y}_{\text{leak}} = \{\mathbf{y}_{i,j} \mid (\mathbf{x}_{i,j}, \mathbf{y}_{i,j}) \in \mathcal{D}_{\text{leak}}\}$. We assume: (i) **Black-box access:** the adversary $\mathcal{A}$ can query the encoder to obtain $\mathbf{z} = F(\mathbf{x}_{1:r})$ for chosen $\mathbf{x}_{1:r}$; (ii) **Partial data leakage:** $\mathcal{A}$ has access to a small, representative sample from the true data domain; and (iii) **Unbounded computation:** $\mathcal{A}$ can construct an optimal reconstruction decoder $D_{\mathcal{A}} : \mathcal{Z} \to \bigcup_{S \in \mathcal{P}(\mathcal{T})} \mathcal{X}_S$. Given a reconstruction loss $\mathcal{L}_{\text{rec}}$ (e.g., mean-squared error), the adversary minimizes (2).

$$\min_{D_{\mathcal{A}}} \; \mathbb{E}_{\mathbf{x}_{1:r} \sim \mathcal{X}_{\text{leak}}} [\mathcal{L}_{\text{rec}}(\mathbf{x}_{1:r}, D_{\mathcal{A}}(F(\mathbf{x}_{1:r})))] \tag{2}$$

3.2.3 Structured noise injection and goals. The spectrum manager deploys a *Structured Noise Injection Mechanism* with user-specified noise budget $\beta \geq 0$, such that

$$\widehat{\mathbf{z}} = M_\beta\big(F(\mathbf{x}_{1:r}), U\big) \quad U \sim \mathrm{P}_U \tag{3}$$

where U is an internal random seed and M_β is a noise injection mechanism (described in detail in § 4.2) parameterized by β. The mechanism must balance two requirements for any chosen task subset $S \subseteq \mathcal{P}(\mathcal{T})$:

1. **Utility.** The released features $\widehat{\mathbf{z}}$ retain sufficient task-relevant information so that there exist task heads $\{G_i\}_{T_i \in S}$ yielding low predictive error on their respective tasks.
2. **Privacy.** Let $D^\star_{\mathcal{A}}$ be an optimal adversarial decoder that attains the minimum in (2). For a given privacy threshold $\xi > 0$, the released features must satisfy (4).

$$\mathbb{E}_{\mathbf{x}_{1:r} \sim \mathcal{X}_{\text{leak}}} \left[\mathcal{L}_{\text{rec}}\big(\mathbf{x}_{1:r}, D^\star_{\mathcal{A}}\big(M_\beta(F(\mathbf{x}_{1:r}), U)\big)\big)\right] \geq \xi \tag{4}$$

Condition (4) describes the notion of privacy used in this work [2]. Note that privacy improves as the expected reconstruction accuracy of any adversary decreases (i.e., reconstruction loss increases), while (3) makes explicit the role of the budget β in trading off utility and privacy. In our setting, we realize M_β as a Gaussian mechanism that perturbs the data with additive noise $U \sim \mathcal{N}(0, \Sigma_n)$, where Σ_n is parameterized using the Fisher Information Matrix (FIM), as detailed in § 4.2.1.

4 Detailed Feature Generation Approach

We first motivate our approach using an information-theoretic reconstruction bound (§ 4.1), then introduce a task-aware anisotropic noise mechanism guided by Fisher directions (§ 4.2). We also interpret the noise budget β via a fixed-budget SNR view (§ 4.2.3) and conclude with a theorem that lower-bounds any adversary's reconstruction error under our structured noise injection.

In the rest of the analysis, $H(\cdot)$ denotes differential entropy, and $I(\cdot;\cdot)$ denotes mutual information. For an input tuple $\mathbf{x}_{1:r} \in \mathbb{R}^d$ (concatenating r components into ambient dimension d), we write $F : \mathbb{R}^d \to \mathbb{R}^k$ for a deterministic encoder producing a latent representation $\mathbf{z} = F(\mathbf{x}_{1:r})$. Noisy representations are $\hat{\mathbf{z}} = \mathbf{z} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_N)$, where $\Sigma_N \succ 0$ is the noise covariance. Unless stated otherwise, expectations are over the joint data–noise distribution.

4.1 Recovering Inputs from Latent Representations

Our objective is to increase the difficulty of reconstructing $\mathbf{x}_{1:r}$ from $\hat{\mathbf{z}}$. Let ξ be the minimum reconstruction error attained by the optimal adversarial decoder $D^\star_{\mathcal{A}}$ as defined in (4). The following lemma lower-bounds ξ via conditional entropy [30].

[2] We note that our scope in the current work is limited to this specific notion of privacy which is well-matched to the use cases of interest. There are many other notions of privacy: our approach is closer to Statistical Disclosure Limitation (SDL)—or disclosure control methods widely used in industry and by statistical agencies for decades [23], rather than differential privacy, which is more popular in computer science theory but has seen relatively less large scale deployment in industry. Exploring variants that match with alternative definitions of privacy is left for future work

LEMMA 1 (MINIMAL RECONSTRUCTION ERROR). *ξ is bounded by the minimum reconstruction error attained by the optimal adversarial decoder $D^{\star}_{\mathcal{A}}$ as in* (5).

$$\xi \;\geq\; \frac{1}{2\pi e}\,\exp\!\left(\frac{2}{d}\,H(\mathbf{x}_{1:r}\mid\hat{\mathbf{z}})\right) \tag{5}$$

Lemma 1 implies that larger $H(\mathbf{x}_{1:r}\,|\,\hat{\mathbf{z}})$ hinders reconstruction. Given $H(\mathbf{x}_{1:r}\mid\hat{\mathbf{z}}) = H(\mathbf{x}_{1:r}) - I(\mathbf{x}_{1:r};\hat{\mathbf{z}})$ where, $H(\mathbf{x}_{1:r})$ is fixed by the data distribution, our goal is to then *minimize* the mutual information $I(\mathbf{x}_{1:r};\hat{\mathbf{z}})$. Note that, Our processing pipeline forms a Markov chain $\mathbf{x}_{1:r} \to \mathbf{z} \to \hat{\mathbf{z}}$ following the data processing inequality $I(\mathbf{x}_{1:r};\hat{\mathbf{z}}) \;\leq\; I(\mathbf{z};\hat{\mathbf{z}})$. Therefore, it suffices to control $I(\mathbf{z};\hat{\mathbf{z}})$.

For an additive Gaussian noise distortion $\hat{\mathbf{z}} = \mathbf{z} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0},\Sigma_N)$ and $\Sigma_z = \mathrm{Cov}(\mathbf{z})$, we have the following standard bound.

$$I(\mathbf{z};\hat{\mathbf{z}}) \;\leq\; \frac{1}{2}\log\det\!\big(\mathbb{I}_k + \Sigma_z\Sigma_N^{-1}\big) \tag{6}$$

with equality when $\mathbf{z}$ is Gaussian [7]. Intuitively, increasing noise power (larger Σ_N) reduces this upper bound, which then increases the lower bound in lemma 1. However, indiscriminately increasing noise power degrades downstream task performance. This motivates a *task-aware, anisotropic* design that preserves high-information directions for the tasks while placing more noise elsewhere.

4.2 Task-Aware Anisotropic Noise Injection

Recall from § 3.2 that $\mathcal{T}$ is the set of downstream tasks and $S \in \mathcal{P}(\mathcal{T})$ is any subset of tasks sharing an encoder F that maps inputs $\mathbf{x}_{1:r} \in \mathcal{X}_S \subseteq \mathbb{R}^d$ to a joint latent $\mathbf{z} = F(\mathbf{x}_{1:r}) \in \mathbb{R}^k$. Each task $i \in S$ is equipped with a task head $G_i : \mathbb{R}^k \to \mathcal{Y}_i$ producing $\hat{\mathbf{y}}_i = G_i(\hat{\mathbf{z}})$ and a loss $\mathcal{L}_i(\hat{\mathbf{y}}_i, \mathbf{y}_i)$.

4.2.1 Fisher directions for task importance. We quantify task sensitivity in latent space via the (empirical) Fisher Information Matrix (FIM) [13] with respect to $\mathbf{z}$, FIM characterizes the sensitivity of a model's likelihood function, or equivalently its loss, to small perturbations in latent representations $\mathbf{z}$. More formally, for a sample $(\mathbf{x},\mathbf{y}_i)$ and its noisy latent $\hat{\mathbf{z}}$, define the gradient $g_i = \nabla_{\mathbf{z}}\mathcal{L}_i(\hat{\mathbf{y}}_i,\mathbf{y}_i) \in \mathbb{R}^k$.[3] The per-sample Fisher contribution is $g_i g_i^\top$, and the empirical FIM for task i over N samples is computed as $\hat{F}_i = \frac{1}{N}\sum_{j=1}^{N} g_{i,j}\, g_{i,j}^\top \in \mathbb{R}^{k\times k}$. Large eigenvalues of $\hat{F}_i$ indicate directions where small latent perturbations strongly affect task i's loss; small eigenvalues indicate task-insensitive directions. The Average FIM, which captures important features across tasks, is then calculated by simply summing the individual FIMs as $\bar{F} = \sum_{i\in S}\hat{F}_i = V\Lambda V^\top, \quad \Lambda = \mathrm{diag}(\lambda_1,\ldots,\lambda_k)$. Important latent directions can then be obtained from the eigenvectors $\{v_1,\ldots,v_k\}$ of the empirical Fisher matrix $\bar{F}$, with their relative importance quantified by the corresponding eigenvalues $\{\lambda_1,\ldots,\lambda_k\}$.

4.2.2 Designing the noise covariance. Given a total noise variance budget $\beta > 0$ (i.e., the total noise power represented as the Trace of the covariance matrix given by $\mathrm{Tr}(\Sigma_N) = \beta$) and a regularizer $\gamma > 0$ for numerical stability and anisotropy control, we allocate variance inversely to task importance along the eigen-directions of $\bar{F}$ as shown in (7).

$$\sigma_i^2 \;=\; c\cdot\frac{1}{\lambda_i+\gamma}, \quad c \;=\; \beta\left(\sum_{j=1}^{k}\frac{1}{\lambda_j+\gamma}\right)^{-1} \tag{7}$$

$$\Sigma_N \;=\; V\,\mathrm{diag}(\sigma_1^2,\ldots,\sigma_k^2)\,V^\top \tag{8}$$

The resulting noise covariance is shown in (8). which concentrates noise along task-insensitive directions (small λ_i) while preserving task-critical ones (large λ_i). As $\gamma\to\infty$, (7) approaches isotropic noise. This structured noise is then injected into the normalized latent representation (see lines 6 - 9 in Algorithm 1).

Algorithm 1 Task-Aware Anisotropic Noise Injection

Require: Input $\mathbf{x}$; encoder F with $\mathbf{z} = F(\mathbf{x}) \in \mathbb{R}^k$; budget $\beta > 0$; eigenpairs (V,Λ) of $\bar{F}$; regularizer $\gamma > 0$; small stabilizer $\tau > 0$.
Ensure: Noisy latent $\tilde{\mathbf{z}}$.

1: **procedure** INJECTANISOTROPICNOISE($\mathbf{z},\beta,V,\Lambda,\gamma$)
2: $\quad z_{\text{norm}} \leftarrow \|\mathbf{z}\|_2$
3: $\quad \mathbf{u} \leftarrow \mathbf{z}/(z_{\text{norm}}+\tau)$ ▷ Normalize to unit sphere
4: $\quad c \leftarrow \beta\left(\sum_{j=1}^{k}\frac{1}{\lambda_j+\gamma}\right)^{-1}$
5: $\quad \sigma_i^2 \leftarrow c/(\lambda_i+\gamma)$ for $i = 1,\ldots,k$
6: $\quad \Sigma_N \leftarrow V\,\mathrm{diag}(\sigma_1^2,\ldots,\sigma_k^2)\,V^\top$
7: $\quad$ Sample $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0},\Sigma_N)$
8: $\quad \tilde{\mathbf{u}} \leftarrow \mathbf{u} + \boldsymbol{\eta}$
9: $\quad$ **return** $\tilde{\mathbf{z}} \leftarrow \tilde{\mathbf{u}}\cdot z_{\text{norm}}$
10: **end procedure**

4.2.3 Signal-to-noise ratio under a fixed budget. Normalizing $\mathbf{z}$ to $\mathbf{u}$ fixes signal power $\|\mathbf{u}\|_2^2 = 1$ (see line 3 of Alg. 1). The injected noise $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0},\Sigma_N)$ has power $\mathbb{E}\|\boldsymbol{\eta}\|_2^2 = \mathrm{Tr}(\Sigma_N) = \sum_{i=1}^{k}\sigma_i^2 = \beta$ by (7). Thus the normalized-space SNR is $1/\beta$, independent of the sample. Rescaling by z_{norm} restores the original magnitude of $\mathbf{z}$ (see line 9). Throughout, we refer to the *noise level* as the budget β. The complete workflow of the structure noise injection is described in Algorithm 1.

THEOREM 1 (TASK-AWARE ANISOTROPIC NOISE AND RECONSTRUCTION ERROR). *Let $\mathbf{x}_{1:r} \to \mathbf{z} \to \hat{\mathbf{z}}$ with $\mathbf{z} = F(\mathbf{x}_{1:r}) \in \mathbb{R}^k$ and $\hat{\mathbf{z}} = \mathbf{z}+\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0},\Sigma_N)$, $\Sigma_N \succ 0$. For tasks $\{T_i\}_{i=1}^n$ with heads $\hat{\mathbf{y}}_i = G_i(\hat{\mathbf{z}})$ and losses $\mathcal{L}_i$, define empirical FIM $\hat{F}_i$ as in § 4.2 and aggregate $\bar{F} = \sum_{i=1}^{n}\hat{F}_i = V\Lambda V^\top$.*

Construct Σ_N via (7)–(8) for a budget $\beta > 0$ and regularizer $\gamma > 0$, so that $\mathrm{Tr}(\Sigma_N) = \beta$. Let $\Sigma_z = \mathrm{Cov}(\mathbf{z})$ and $H(\mathbf{x}_{1:r})$ denote the differential entropy of the input.

Then for any (unbounded) adversarial decoder $D_{\mathcal{A}} : \mathbb{R}^k \to \mathbb{R}^d$, the (per-coordinate) MSE $\xi = \frac{1}{d}\,\mathbb{E}\big[\|\mathbf{x}_{1:r} - D_{\mathcal{A}}(\hat{\mathbf{z}})\|_2^2\big]$ is bounded as denoted by (9).

$$\xi \;\geq\; \frac{1}{2\pi e}\,\exp\!\left(\frac{2}{d}H(\mathbf{x}_{1:r}) - \frac{1}{d}\log\det\!\big(\mathbb{I}_k + \Sigma_z\Sigma_N^{-1}\big)\right) \tag{9}$$

5 Use-cases for Shared Feature Generation

In this section, we describe the problem setup with three wireless communication tasks that leverage ML, the dataset that we intend to use, and the model architectures that we propose to train for classification or regression outputs.

[3]Gradients are evaluated at the clean latent $\mathbf{z}$ unless noted.

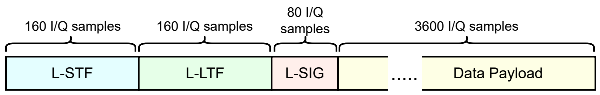

Figure 2: IEEE 802.11 Non-HT WiFi packet structure: preamble fields L-STF, L-LTF, L-SIG, and data payload.

5.1 Dataset and Task Description

We use a publicly available OTA WiFi dataset [17] collected indoors. 16 USRPs transmit IEEE 802.11 Non-HT WiFi frames (QPSK, 3/4 coding, 2.4 GHz, 5 MHz bandwidth) at distances 2–62 ft; a B210 USRP records all activity. Each extracted packet (Fig. 2) has a preamble with L-STF (160 I/Q), L-LTF (160 I/Q), and L-SIG (80 I/Q), followed by a 3600-sample data payload. We design three tasks based on these fields.

5.1.1 Task 1. RF fingerprinting. RF fingerprinting is a physical layer device authentication technique that can detect subtle distortions in the received waveform that are caused by the transmitter hardware imperfections, and consequently identify the transmitter. We select our wireless task 1 as RF fingerprinting and formulate it as a classification task, where the inputs are I/Q slices of the received packet, and the outputs are class indexes for the 16 different transmitters in the dataset (i.e., 0 to 15).

5.1.2 Task 2. Coarse CFO estimation. Coarse CFO estimation is the second processing step in the receiver chain that happens immediately after packet detection (a.k.a., coarse time synchronization). In the WiFi OFDM receiver, coarse CFO estimation is done using the first field of the preamble, i.e., L-STF. We design our second wireless task to be coarse CFO estimation, and we formulate it as a regression problem where the inputs are L-STF sequences and the outputs are the coarse CFO values that are estimated using the traditional algorithm implemented in MATLAB [26].

5.1.3 Task 3. Channel estimation. Channel estimation is an integral part of the wireless receiver chain, that is traditionally done using the L-LTF sequence. We design our third wireless task to be channel estimation with L-LTF as input and a complex-valued vector of length 52 as output. The complex-valued estimated channel is obtained through traditional channel estimation in MATLAB.

5.2 Encoder Description

The encoder architecture consists of three main components: task-specific projection layers, a shared encoder, and task-specific heads.

Projection Layers. For each task $T_i \in \mathcal{T}$, we use a lightweight projection layer to map inputs $\mathbf{x}_i \in \mathcal{X}_i$ into a common, fixed-dimensional space. This allows the shared encoder to process inputs from different tasks uniformly. Each projection layer is a single 1D convolutional layer. The hidden dimension of the projection layer is set to 512, and it maps all inputs to a common sequence length of 512.

Fusion Mechanism. Following the projection stage, feature maps from tasks $S \subseteq \mathcal{T}$ are combined via summation, $\mathbf{x}_{\text{fused}} = \sum_{T_i \in S} \text{proj}_i(\mathbf{x}_i)$, or depth-wise concatenation, $\mathbf{x}_{\text{fused}} = \text{concat}(\{\text{proj}_i(\mathbf{x}_i)\}_{T_i \in S})$.

Shared Encoder. Figure 3 shows the architectures of shared encoder and task-specific heads. The shared encoder is the main component for feature extraction and is designed to learn a shared representation from the fused projections. The architecture consists of 2 sequential 1D convolutional blocks. The output is then flattened and passed through a linear layer to produce a latent representation with an output dimension of $d_2 = 256$. Given the complex-valued nature of the data (I/Q channels), this results in a final real-valued latent vector $\mathbf{z} \in \mathbb{R}^{512}$ which is reshaped to 2×256. A dropout rate of 0.1 is applied throughout the encoder.

Task Heads. The latent representation $\mathbf{z}$ is fed into separate, task-specific heads, $\{G_i\}_{i=1}^{N}$, each with a hidden dimension of 256. The RF Fingerprinting head is a classifier that processes the feature vector through linear layers and dropout, producing a 16-dimensional output for training with a Cross-Entropy loss. The CFO Estimation head is a regression model that uses linear layers and dropout to predict a single continuous value, trained by minimizing Mean Squared Error (MSE). Similarly trained with MSE, the Channel Estimation head is a regression model that predicts 52 channel coefficients.

5.2.1 Encoder Training. The overall objective is to find the optimal parameters for the encoder F and task heads $\{G_i \mid T_i \in S\}$ that minimize a joint weighted loss as in (10).

$$\min_{F,\{G_i \mid T_i \in S\}} \sum_{T_i \in S} w_i \mathcal{L}_i(G_i(F(\mathbf{x}_{\text{fused}})), \mathbf{y}_i) \tag{10}$$

Here, $\mathcal{L}_i$ is the task-specific loss function, $\mathbf{y}_i$ is the ground-truth label, and w_i are task weights, which we set to 1. The training process begins by passing inputs through their respective projection layers, denoted $\text{proj}_i(\cdot)$. The resulting features are combined into a single fused input, $\mathbf{x}_{\text{fused}}$, using either element-wise summation or channel-wise concatenation. For the RF fingerprinting task, which involves longer signal sequences, we consider two data processing strategies: (i) a *variable-RF* setting, where random subsampling is performed on-the-fly, and (ii) a *fixed-RF* setting, where a predetermined segment of length 1024 is consistently selected from each signal. Given a raw signal sample $\mathbf{x}_{\text{rf}} \in \mathcal{X}_{\text{rf}}$ of length L, a training instance $\mathbf{x}'_{\text{rf}}$ of length 1024 is constructed as follows. In the variable-RF setting, a starting index k is sampled uniformly at random from the interval $[401, L-1023]$, and the corresponding subsequence is extracted. This procedure ensures that, across training epochs, the model is exposed to diverse segments of the signal while systematically excluding the initial 400 samples, which may contain transient effects. In contrast, in the fixed-RF setting, a single starting index is predetermined for each signal, and the same 1024-length segment is used consistently throughout training.

5.3 Adversary Description

Decoder Architecture. The architecture of adversary is shown in figure 4. The adversary's decoder, $D_{\mathcal{A}}$, is designed to invert the encoding process, reconstructing the original input signals from the latent representation $\mathbf{z}$. The architecture mirrors the encoder's structure in reverse, employing a shared backbone and task-specific heads. The shared backbone first processes the flattened 512-dimensional latent vector through a series of linear layers with LeakyReLU activations and dropout, progressively increasing its dimensionality. This shared representation is then channeled into one of three task-specific heads, each consisting of 1D transposed

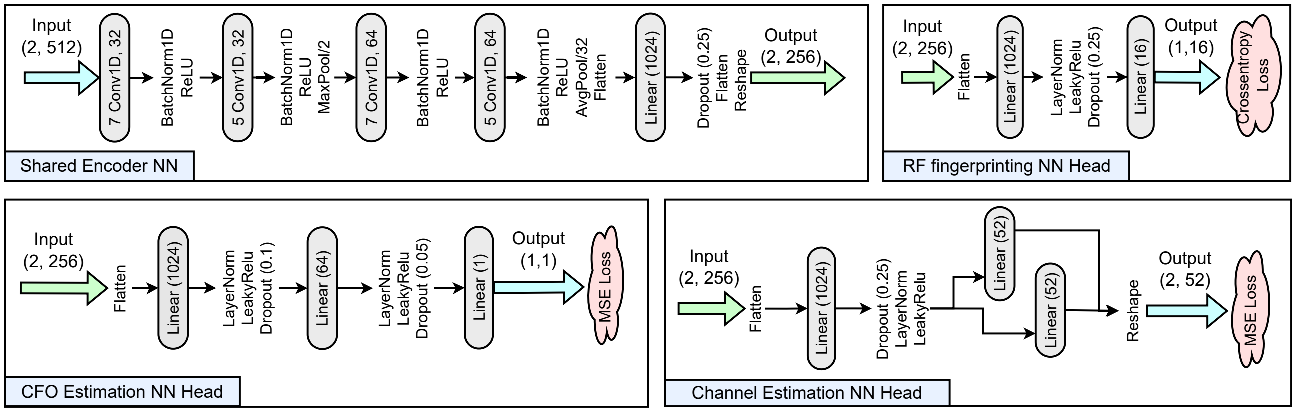

Figure 3: Shared encoder architecture and task-specific heads for RF fingerprinting, CFO estimation, and Channel estimation.

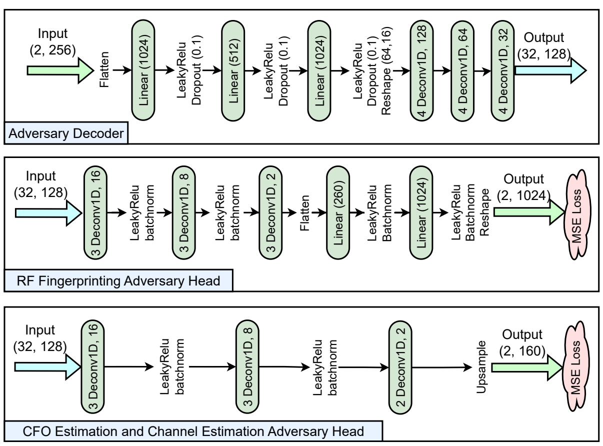

Figure 4: Adversary decoder architecture: shared backbone and task-specific heads for RF, CFO, and Channel reconstruction.

convolutional layers to match the original input dimensions. The decoder is trained to minimize (2) using MSE as the reconstruction loss.

6 Performance Evaluation

This section presents the experiment setup and evaluation metrics (§ 6), as well as the empirical findings from our framework. Our evaluations address: (i) Does joint learning improve utility? (§ 6.2); (ii) Does it enhance privacy? (§ 6.3); (iii) Does the encoder generalize to unseen tasks? (§ 6.8); (iv) How does the privacy–utility tradeoff compare with the isotropic baseline? (§ 6.4); (v) Is the mechanism robust to varying data leakage? (§ 6.5); (vi) How does our method compare with baselines? (§ 6.7); (viii) Does random slicing amplify privacy? (§ 6.9).

6.1 Setup and Evaluation Metrics

6.1.1 Experiment setup. We consider encoders pretrained on all non-empty subsets $S_i \subseteq \mathcal{T}$, where $\mathcal{T} = \{\text{RF}, \text{CFO}, \text{Channel}\}$ corresponds to the tasks in § 5.1.1–5.1.3.

For each pretrained encoder, we construct an adversarial attack under the assumption that a fraction of the input space $\mathcal{X}$, denoted as the *leakage fraction*, is exposed together with its noise-corrupted latent representations. The corruption noise level (β) is varied in the range $[5, 20]$, with the relation of β to signal-to-noise ratio given in § 4.2.3. As a baseline, we consider isotropic noise injection at the same noise levels. Furthermore, we vary the parameter γ as described in § 4.2.2 as a control knob to tune the degree of *anisotropy* in the injected noise. The different values of these parameters are summarized in Table 1. In addition to these parameters, we also compare the fusion mechanisms summation and concatenation as discussed in § 5.2 used to combine task-specific representations.

Table 1: parameter grid explored for evaluation.

Parameter	Values
Noise type	None, Isotropic, Anisotropic
Noise levels	0 (none), {5, 10, 15, 20} (for isotropic/anisotropic)
Leakage fraction	{0.02, 0.04, 0.06, 0.08, 0.1, 0.5, 1.0}
γ	{10, 1, 0.1, 0.01, 0.001, 0.0001} (anisotropic only)

The encoder is trained for up to 300 epochs using Adam (lr = 10^{-4}), cosine annealing, and early stopping with patience 10. The decoder is trained for up to 40 epochs on $\mathcal{D}_{\text{leak}}$ using AdamW (lr = 10^{-3}), ReduceLROnPlateau, early stopping with patience 10. All results are reported under the RF variable setting unless otherwise stated.

6.1.2 Evaluation metrics.

Privacy. Privacy is quantified as *Adversary-NMSE*: the normalized mean squared error of the adversary's reconstruction $D_{\mathcal{A}}(\hat{\mathbf{z}})$ relative to $\mathbf{x}$, as defined in (2). Higher adversary-NMSE indicates stronger privacy.

Utility. For RF Fingerprinting (16-class classification), we report test accuracy. For CFO and Channel estimation, we report the R^2 score ($R^2 = 1 - \sum \|\hat{y} - y\|^2 / \sum \|y - \bar{y}\|^2$), where $\bar{y}$ is the empirical mean of ground-truth values.

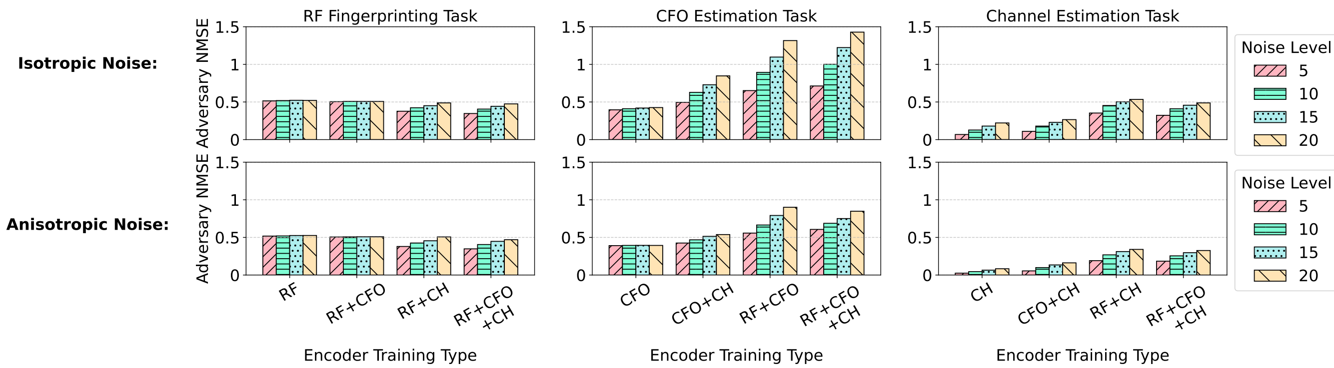

Figure 5: Adversary-NMSE for RF, CFO, and Channel reconstruction tasks under isotropic and anisotropic noise at levels 5–20, comparing single- and joint-task encoders. Joint learning increases privacy for CFO and Channel; RF remains near NMSE ≈0.5.

6.2 Joint Learning Improves Utility

Table 2: Comparison of different strategies for constructing multi-task inputs. Summation yields consistently better results compared to concatenation.

Strategy	Main Task Metric	Single Task	Two-task +RF	Two-task +CFO	Two-task +Channel	Three Task
Summation	RF Accuracy	0.85	-	0.96	0.94	**0.97**
	CFO R^2	0.64	0.56	-	**0.69**	0.65
	Ch. R^2	**0.99**	0.98	0.98	-	0.98
Depth-wise Concatenation	RF Accuracy	0.83	-	0.92	0.79	**0.94**
	CFO R^2	0.61	0.56	-	**0.64**	0.57
	Ch. R^2	**0.99**	0.97	0.98	-	0.97

In the joint learning setting, we explore two fusion mechanisms, summation and depthwise concatenation, to combine projected inputs before passing them to the shared encoder.

As shown in Table 2, joint learning generally improves utility across tasks. RF Fingerprinting benefits the most, with accuracy rising from 85% to 97% under summation in the three-task setting. CFO Estimation also improves modestly (R^2: 0.64 to 0.65), with its highest gains when paired with Channel Estimation. Channel Estimation itself sees negligible change, as it was already near-perfect (99%) in the single-task setting.

Comparing fusion strategies, summation consistently outperforms depthwise concatenation. Both yield a 12% gain in RF Fingerprinting, but summation achieves 97% vs. 94% with concatenation. Moreover, depthwise concatenation degrades CFO Estimation (R^2: 0.61 to 0.57), whereas summation improves it. We therefore adopt summation as the default fusion strategy for the remainder of our analysis.

For the summation fusion strategy, Fig. 6 shows utility versus distance for all tasks under joint encoder training. The jointly trained encoder (green curves) consistently outperforms single-task baselines in RF fingerprinting and channel estimation, with accuracy degrading only at longer distances. This demonstrates the advantage of shared representations for RF tasks. For CFO estimation, the joint CFO+RF model maintains high accuracy averaged across all distances (see Table 2).

6.3 Joint Learning Improves Privacy Compared to Single-Task Learning

Figure 5 shows adversarial reconstruction performance for different single and joint task learning experiments, under isotropic and anisotropic noise. For the RF task, adversary-NMSE remains close to 0.5 across all configurations, indicating that RF signals are inherently difficult to reconstruct regardless of joint training. We revisit this property in §6.9. For the CFO task, joint learning leads to substantial privacy gains. Under isotropic noise, adversary-NMSE rises from 0.40 in the single-task setting to 0.71 with all three tasks trained jointly, reaching as high as 1.43 at noise level 20. Anisotropic noise shows a similar pattern: adversary-NMSE increases from 0.39 (CFO only) to 0.61 (RF+CFO+CH) at noise level 5, and up to 0.85 at noise level 20. Two-task learning (CFO+CH) already improves privacy meaningfully (e.g., 0.43 at noise level 5), but full three-task learning amplifies this effect. For the Channel task, adversary-NMSE also increases with joint training. Under isotropic noise, adversary-NMSE grows from 0.07 in the single-task setting to 0.32 (RF+CFO+CH) at noise level 5 and 0.49 at noise level 20. Under anisotropic noise, the trend is similar but slightly weaker: adversary-NMSE rises from 0.02 (CH only) to 0.18 (RF+CFO+CH) at noise level 5 and 0.33 at noise level 20. However, as we will see in the next sections, this is compensated by a smaller degradation in utility.

6.4 Anisotropic Noise Addition Improves Privacy-Utility Tradeoff

We consider the joint three-task setting (RF + CFO + CH) to compare the privacy–utility tradeoff of our proposed anisotropic noise mechanism against the isotropic baseline. As shown in Figure 7, the anisotropic approach consistently outperforms the isotropic baseline, with performance curves positioned above and to the right, indicating higher utility at equal privacy, and stronger privacy at equal utility. The improvement is most evident in the Channel Estimation task. At a low noise level of $\beta = 5$, the isotropic baseline yields an adversary-NMSE of 0.21 with a utility of only 0.35 (R^2). In comparison, our method achieves similar privacy with NMSE = 0.20, but higher utility of R^2=0.63. This gap widens further at

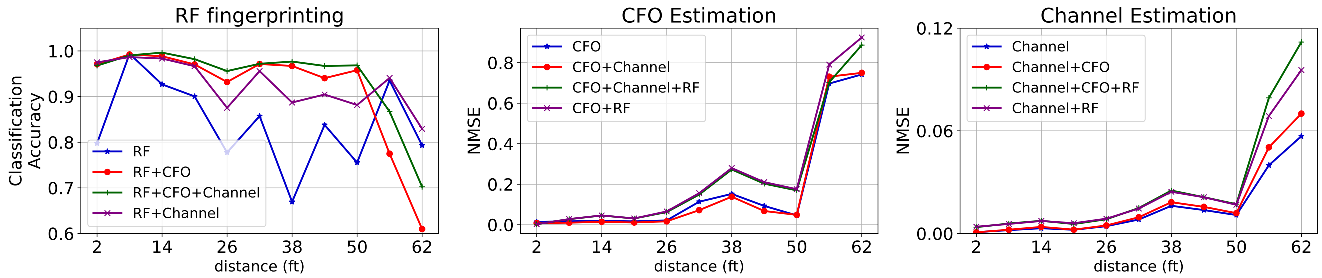

Figure 6: Evaluation of detection performance for all three tasks against distance (in ft) for the joint encoder training, with summation as fusion strategy. Jointly trained encoder (green) outperforms single-task models, most evident in RF Fingerprinting and Channel estimation. In RF fingerprinting, joint learning mitigates mid-range accuracy degradation. For CFO Estimation, RF+CFO achieves best accuracy across all distances. This partly explains the trends observed in Table 2.

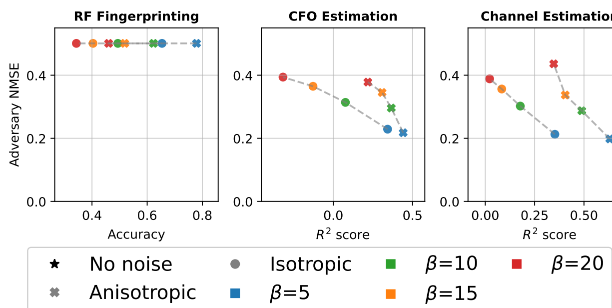

Figure 7: Privacy–utility tradeoff in the joint three-task model for noise levels 5–20. Anisotropic noise lies above and right of isotropic, indicating better utility at equal privacy and stronger privacy at equal utility.

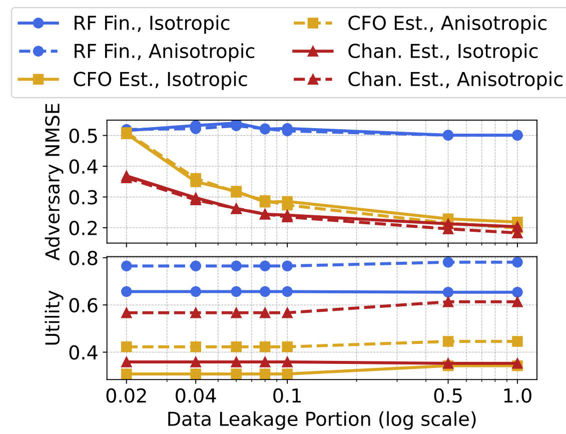

Figure 8: Effect of adversary data access on privacy (top) and utility (bottom). Anisotropic noise maintains higher utility at comparable privacy as leakage increases.

higher noise levels. At noise level 20, the isotropic baseline degrades to near-zero utility ($R^2 = 0.02$), while our method preserves significantly higher utility ($R^2 = 0.35$), with comparable privacy as measured by adversary NMSE (0.44 vs. 0.39 for anisotropic).

A similar trend holds for CFO Estimation. At noise level 20, the baseline utility degrades below 0 (i.e., $R^2 = -0.32$), while our method preserves a positive $R^2 = 0.22$ with negligible change in privacy (NMSE = 0.38 vs. 0.39). At lower noise levels, such as $\beta = 5$, our method again delivers a stronger tradeoff, achieving a higher utility of $R^2 = 0.44$ (vs. 0.34) with similar privacy (NMSE = 0.22 vs. 0.23).

For RF Fingerprinting, the adversary's NMSE remains close to 0.5 for both mechanisms across all noise levels, suggesting that reconstructing private features in this task is inherently difficult (more on this in § 6.9).

6.5 Impact of Data Leakage on Adversary Reconstruction Error

Figure 8 shows the effect of increasing adversary data access on privacy and utility. As leakage increases, adversary-NMSE decreases for both noise types (intuitively consistent), but anisotropic noise maintains higher utility throughout. For CFO Estimation, utility is 0.42 vs. 0.30 at 2% leakage and 0.44 vs. 0.34 at full access. For Channel Estimation, at 2% leakage, adversary-NMSE is comparable (0.36 vs. 0.37) while utility is substantially higher (0.57 vs. 0.36, respectively). Even at 100% leakage, anisotropic noise retains a utility of 0.61 compared to 0.35 for isotropic noise for a similar privacy guarantee. For RF Fingerprinting, adversary-NMSE remains around 0.5 for both noise types across all leakage levels, consistent with prior observations. These results confirm that the advantage of anisotropic noise persists even as the adversary becomes stronger.

6.6 Parameter γ : Isotropy vs. Anisotropy

The parameter γ interpolates between anisotropic and isotropic noise regimes. As γ increases, adversary performance under anisotropic noise approaches the isotropic case; for $\gamma \geq 1$, the two are nearly indistinguishable. At $\gamma = 10^{-4}$, the anisotropic characteristics are amplified, with channel estimation adversary-NMSE/Utility rising from ~0.3 to ~0.8 at $\gamma = 10$, converging to the isotropic baseline.

6.7 Comparison with Baselines

Renyi-DP baseline. Standard DP does not directly apply to *per-sample* instance encodings. Similar to [18], We adapt it by applying

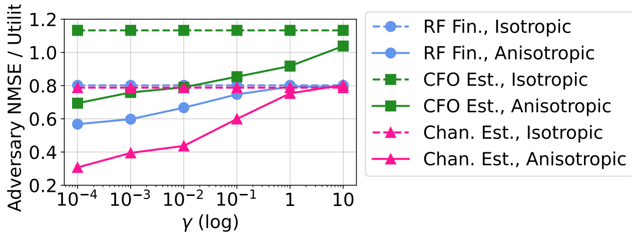

Figure 9: Sensitivity of adversary reconstruction error (i.e., adversary-NMSE) to Isotropy to Anisotropy variation, realized using γ parameter.

Table 3: Utility at matched privacy (adversary recon. MSE matched per task). RDP σ is chosen per task so its recon. MSE equals ours. No-noise baseline shows utility without any privacy mechanism.

	Privacy (MSE↑)		Utility		
Task	Ours	RDP	No noise	Ours	RDP
RF (Acc.)	0.501	0.501	0.944	**0.519**	0.073
CFO (R^2)	0.345	0.345	0.601	**0.308**	−0.141
Channel (R^2)	0.337	0.337	0.981	**0.405**	−0.075

the Gaussian mechanism at the instance level, clip $\|\mathbf{z}\|_2 \leq 1$ and add $\mathcal{N}(\mathbf{0}, \sigma^2 I)$, satisfying $(2, \epsilon)$-RDP per release. We calibrate σ per task to match our method's adversary reconstruction MSE. This requires $\sigma \approx 39$–46, which collapses utility entirely (Table 3). RF accuracy falls to 0.073 (near the 0.0625 random baseline), far below the no-noise baseline of 0.944, and regression performance goes negative for CFO (−0.141) and Channel (−0.075). This demonstrates that achieving instance-level DP via isotropic noise is destructive in MTL setups.

NeuraCrypt. NeuraCrypt applies a frozen random CNN encoder with per-sample patch shuffling; to adapt it to our setting, we convert raw I/Q samples to their 2D STFT representation and treat it as an image input to the encoder. It achieves reconstruction MSE of 0.497, 0.039, and 0.043 on RF, CFO, and channel tasks, respectively, with corresponding utility of 0.081 accuracy, $R^2 = 0.037$, and $R^2 = 0.837$. The near-random performance in RF and CFO tasks is likely attributable to patch shuffling, which disrupts the temporal structure required for these tasks; however, channel estimation appears more resilient. Our method demonstrates a more favorable privacy-utility tradeoff compared to the Neuracrypt baseline.

Training-time methods. While invertibility-penalized [18] and adversarial methods require training-time or architecture-level modifications, our approach operates strictly post-training. We therefore defer comparisons with such methods.

DP connection. Our primary privacy guarantee is *reconstruction hardness*: Theorem 1 lower-bounds the MSE of any adversary attempting to recover $\mathbf{x}_{1:r}$ from $\hat{\mathbf{z}}$, and this is the guarantee evaluated throughout §6.3–6.5. As an *auxiliary* result, we note that our noise construction also admits a formal (ϵ, δ)-DP bound, a conceptually distinct guarantee concerned with indistinguishability of adjacent inputs rather than reconstruction difficulty. Clipping $\|\mathbf{z}\|_2 \leq \eta_{\max}$ and applying the standard Gaussian mechanism, we have, $\epsilon \leq \frac{2\eta_{\max}\sqrt{2\ln(1.25/\delta)}}{\sigma}$ [10]. In our anisotropic setting, the worst-case ϵ is determined by the direction of minimum noise (largest FIM eigenvalue $\lambda_{\max}$). Substituting $\sigma_{\min} = \sqrt{c/(\lambda_{\max} + \gamma)}$ and taking the worst-case $\lambda_{\max} \approx 1$, $\lambda_i \approx 0$ for $i > 1$ results in (11).

$$\epsilon \leq \sqrt{\frac{8\eta_{\max}^2 \ln(1.25/\delta)}{\beta}} \sqrt{1 + \frac{(d-1)(1+\gamma)}{\gamma}} \tag{11}$$

The second factor represents the anisotropy overhead: for $\gamma = 1$ it is $\approx \sqrt{2d}$, versus $\sqrt{d}$ for isotropic noise, meaning $\beta_{\text{aniso}} \approx 2\beta_{\text{iso}}$ at equal worst-case ϵ. We stress that DP and reconstruction hardness measure fundamentally different properties; DP quantifies worst-case input indistinguishability, while reconstruction hardness bounds the adversary's recovery error, and neither implies the other in general. The DP bound here is a supplementary formal property of our noise construction, not a substitute for the empirical tradeoff analysis in §6.4.

6.8 Generalization to Unseen Tasks

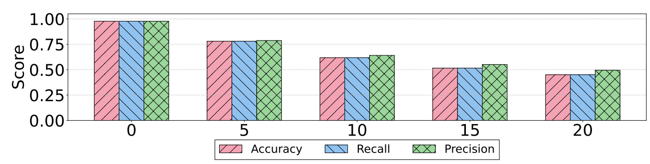

Figure 10: Generalization to the unseen distance estimation task across non-isotropic noise levels. High utility scores persist despite the encoder never being trained on distance estimation.

To assess the generalizability of our learned representations, we evaluate the encoder on a downstream task unseen during training: predicting the physical distance between transmitter and receiver. Using the encoder originally trained on RF fingerprinting, CFO, and channel estimation, we freeze all encoder layers and train a lightweight task head on data collected at varying distances.

As shown in Fig. 10, the model achieves approximately 97% across accuracy, precision, and recall for all distance classes at $\beta = 0$, demonstrating transferability to an unseen related task. Performance degrades gracefully as noise increases, consistent with findings in other tasks. However, we note that one should not expect a similar transfer to tasks that are fundamentally unrelated to the original tasks, as explained in the introduction. This fundamentally happens when the regularization provided by multitask learning is not germane to a highly unrelated task.

6.9 Privacy Amplification Through Random Slicing

We compare two training schemes for RF fingerprinting: Fixed RF, where the encoder receives fixed non-overlapping 1024-length slices of the input, and Variable RF, where a 1024-length segment is randomly subsampled at each training step (see § 5.2). This analysis is motivated by prior observations that subsampling from time-series data can act as a privacy amplification mechanism [16]. With no added noise, both regimes achieve high utility (accuracy ≈ 0.95), yet Fixed RF leaks substantially more information (adversary-NMSE = 0.26) compared to Variable RF (adversary-NMSE = 0.50). This advantage persists across all noise levels: Variable RF consistently maintains adversary-NMSE ≈0.5 while matching Fixed RF utility. One way to interpret this is through the lens of regularization. By presenting only subsampled views of the input during training, Variable RF makes it more difficult for the encoder to memorize any particular training sample $\mathbf{x}$, and hence more difficult for the adversary to decode from its latent representations.

7 Conclusion

In conclusion, we introduced a framework for privacy-aware latent feature sharing in wireless datasets and showed that joint training simultaneously improves utility and privacy. We further established a formal privacy guarantee via a lower bound on adversarial reconstruction error and proposed a task-aware structured noise injection mechanism for multi-task learning settings, that outperforms isotropic baselines in the privacy–utility tradeoff (e.g., sustaining $R^2 = 0.35$ in Channel Estimation versus 0.02 for isotropic baseline at comparable privacy). These results highlight joint representations as a powerful candidate for privacy-preserving wireless data sharing. Future directions include developing privacy-aware foundation models with such formal guarantees on privacy and exploring training-time interventions such as adversarial training for stronger privacy guarantees.

Acknowledgments

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A Appendix

Proof of Theorem 1 . For a fixed noisy representation $\hat{\mathbf{z}} = \hat{\mathbf{z}}_0$, consider reconstructing $\mathbf{x}_{1:r}$ from $\hat{\mathbf{z}}_0$. Among all estimators, the minimum mean-squared error (MMSE) estimator is optimal. Applying the Shannon lower bound [7] to the conditional distribution $p(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}} = \hat{\mathbf{z}}_0)$ gives

$$\frac{1}{d}\mathbb{E}\big[\|\mathbf{x}_{1:r} - D_{\mathcal{A}}(\hat{\mathbf{z}})\|_2^2 \,\big|\, \hat{\mathbf{z}} = \hat{\mathbf{z}}_0\big] \geq \frac{1}{2\pi e}\exp\!\bigg(\frac{2}{d}h(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}} = \hat{\mathbf{z}}_0)\bigg). \tag{12}$$

Taking expectation over $\hat{\mathbf{z}}$ yields

$$\xi \geq \frac{1}{2\pi e}\mathbb{E}_{\hat{\mathbf{z}}}\left[\exp\left(\frac{2}{d}h(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}})\right)\right]. \tag{13}$$

Since the exponential function is convex, Jensen's inequality gives

$$\xi \geq \frac{1}{2\pi e}\exp\left(\frac{2}{d}h(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}})\right). \tag{14}$$

Since $\mathbf{z} = F(\mathbf{x}_{1:r})$ is a deterministic function of the input and the additive noise $\boldsymbol{\epsilon}$ is independent of both $\mathbf{x}_{1:r}$ and $\mathbf{z}$, we have the Markov chain

$$\mathbf{x}_{1:r} \longrightarrow \mathbf{z} \longrightarrow \hat{\mathbf{z}}.$$

Using the identity

$$h(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}}) = H(\mathbf{x}_{1:r}) - I(\mathbf{x}_{1:r}; \hat{\mathbf{z}}),$$

and the data-processing inequality,

$$I(\mathbf{x}_{1:r}; \hat{\mathbf{z}}) \leq I(\mathbf{z}; \hat{\mathbf{z}}),$$

we obtain

$$h(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}}) \geq H(\mathbf{x}_{1:r}) - I(\mathbf{z}; \hat{\mathbf{z}}). \tag{15}$$

Consider, $\hat{\mathbf{z}} = \mathbf{z} + \boldsymbol{\epsilon}$, the mutual information then satisfies

$$I(\mathbf{z}; \hat{\mathbf{z}}) = h(\hat{\mathbf{z}}) - h(\hat{\mathbf{z}} \mid \mathbf{z}).$$

Conditioned on $\mathbf{z}$,

$$\hat{\mathbf{z}} = \mathbf{z} + \boldsymbol{\epsilon},$$

so translation invariance of differential entropy gives

$$h(\hat{\mathbf{z}} \mid \mathbf{z}) = h(\boldsymbol{\epsilon}).$$

Therefore,

$$I(\mathbf{z}; \hat{\mathbf{z}}) = h(\hat{\mathbf{z}}) - h(\boldsymbol{\epsilon}).$$

The covariance of $\hat{\mathbf{z}}$ is

$$\mathrm{Cov}(\hat{\mathbf{z}}) = \Sigma_z + \Sigma_N.$$

Among all distributions with a fixed covariance, the Gaussian distribution has the largest differential entropy [7]. Hence,

$$h(\hat{\mathbf{z}}) \leq \frac{1}{2}\log\left((2\pi e)^k \det(\Sigma_z + \Sigma_N)\right).$$

Since $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_N)$,

$$h(\boldsymbol{\epsilon}) = \frac{1}{2}\log\left((2\pi e)^k \det(\Sigma_N)\right).$$

Subtracting the two expressions gives

$$I(\mathbf{z}; \hat{\mathbf{z}}) \leq \frac{1}{2}\log\frac{\det(\Sigma_z + \Sigma_N)}{\det(\Sigma_N)} \tag{16}$$

$$= \frac{1}{2}\log\det\left(\mathbb{I}_k + \Sigma_z\Sigma_N^{-1}\right). \tag{17}$$

Step 4: Combine the bounds.

Substituting (17) into (15) gives

$$h(\mathbf{x}_{1:r} \mid \hat{\mathbf{z}}) \geq H(\mathbf{x}_{1:r}) - \frac{1}{2}\log\det\left(\mathbb{I}_k + \Sigma_z\Sigma_N^{-1}\right).$$

Substituting this inequality into (14) yields

$$\xi \geq \frac{1}{2\pi e}\exp\left(\frac{2}{d}H(\mathbf{x}_{1:r}) - \frac{1}{d}\log\det\left(\mathbb{I}_k + \Sigma_z\Sigma_N^{-1}\right)\right),$$

which is exactly the desired result. □

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